ROUGE SCORE

Presented by Group 5

SOU Pichchomrong



Our Team

YENG Kheang

KE Manut

SOU Pichchomrong

KUONG Sivhoung



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Introduction

The ROUGE (Recall Oriented Understudy for Gisting Evaluation) score is mainly used to evaluate text summarization and other text generation tasks.

- It measures how much of the important information from the reference text is captured in the generated output.

The final ROUGE score ranges from 0 to 1, where higher values indicate better content coverage and similarity to the reference text.

Measure	Technical explanation	Plain-language explanation
Recall	`M / R`: proportion of reference units recovered by the candidate.	“How much of the reference did the summary cover?”
Precision	`M / C`: proportion of candidate units that match the reference.	“How much of what the system wrote matches the reference?”
F1	`2 × Precision × Recall / (Precision + Recall)`: harmonic mean of precision and recall.	“How well does the summary balance coverage and matching content?”

Types of ROUGE(5)

2.1 ROUGE-N: CONSECUTIVE N-GRAM OVERLAP

What it measures: Matching sequences of **N consecutive token.**

N = 1: ROUGE-1, individual words.

N = 2: ROUGE-2, consecutive word pairs.

N = 3: ROUGE-3, consecutive three-word sequences.

Example :

Reference: the weather is cold outside

Candidate: the weather is cold

2.2 ROUGE-L: Longest common suvsequence

What it measures: A longest common subsequence (LCS)—the longest sequence of matching tokens in the same order, with gaps allowed. Unlike a consecutive phrase, a subsequence can skip intervening words.

Reference: the cat sat on the mat
Candidate: the cat rested on the mat

LCS: the cat on the mat . (length=5)

2.3 ROUGE-Lsum: Summary-level union LCS

Summary-Level LCS overlap using the union of matched reference-token positions across sentence alignments. This differs from treating the entire summary as one global sequence.

2.4 ROUGE-W: Weighted longest common subsequence

What it measures: A weighted LCS that gives more credit to consecutive matching runs than to scattered matches.

Reference: A B C D E F G
Candidate 1: A B C D H I K
Candidate 2: A H B K C I D

Position ROUGE-W as a refinement of ROUGE-L — "LCS finds what matched; W cares how it matched."

2.5 ROUGE-S: Skip-bigram overlap

What it measures: Matching ordered token pairs, allowing gaps between the two tokens.

Reference: the cat sat
Candidate: the cat slept

Reference skip-bigrams:
(the, cat), (the, sat), (cat, sat)

Candidate skip-bigrams:
(the, cat), (the, slept), (cat, slept)

Recall = 1/3
Precision = 1/3
F1 ≈ 0.3333

2.6 ROUGE-SU: Skip-bigrams plus unigrams

What it measures: Skip-bigram overlap plus individual-word overlap. The added unigrams provide credit when texts share words but no ordered pairs.

Reference: A B C
Candidate: C B A

- Shared unigrams: A, B, C → three matches.
- Reference skip-bigrams: AB, AC, BC.
- Candidate skip-bigrams: CB, CA, BA.
- Matching skip-bigrams: zero.

Matching units = 3 unigrams + 0 skip-bigrams = 3
Reference units = 3 unigrams + 3 skip-bigrams = 6
Candidate units = 3 unigrams + 3 skip-bigrams = 6

Recall = Precision = F1 = 3/6 = 0.50

Limitation

Limitation	Explanation
Limited semantic understanding	Cannot recognize similar meanings expressed using different words.
Cannot detect factual errors	May give high scores to sentences that share words but have opposite meanings.
Reference dependence	May penalize valid summaries that use different wording or information.
Limited word-order sensitivity	ROUGE-1 may give high scores even when word order changes the meaning.
Recall favors longer output	Longer summaries may capture more reference words but include unnecessary information.
Does not assess readability	Cannot directly evaluate grammar, coherence, fluency, or logical consistency.
Sensitive to configuration	Results can change depending on tokenization, stemming, and other settings.
Language-dependent preprocessing	English-based tokenization may not work well for languages with different writing systems.

Thank You

